Finite-sample edge-order inference and certified degree comparisons in correlation spanning trees

Abstract

A stable estimated hub of a correlation spanning tree can still have an uncertain population rank. We construct finite-sample-valid edge-order inference using e-values for Kendall tau-a differences. Averaging evidence across observation matchings controls familywise error under IID sampling, without moment assumptions or auxiliary tie randomization. A sample-size-dependent betting fraction ensures consistency for every fixed positive gap. Joint projection combines graphic-minor degree comparisons, cycle-path linear programming, rational dual certificates and degree conservation. Matched simulations compare eight calibrations, five projections, and local versus complete edge families. In separated 52-vertex models with 4096 observations, evidence aggregation identifies the hub in all 100 datasets per model, compared with 67–72% for single-matching McNemar inference; near-tie designs remain unresolved. Gaussian multiplier contrasts offer stronger short-sample resolution but exhibit finite-size error inflation in some null cells. A separate paired audit examines Pearson-bootstrap studentization and resample count. For 4177 Abalone specimens, exact evidence aggregation restricts whole-weight rank to the top three and determines peripheral leaf degrees under an explicit IID working model. Polish equity results provide a separate sensitivity study for dependent returns. The framework distinguishes informative population restrictions from estimated-tree stability while making calibration and computational costs inspectable.

Keywords: Kendall tau-a; e-values; simultaneous inference; spanning tree; U-statistics; degree rank.

1 Introduction

Which measured variable is the hub of a population correlation tree? This question arises when a network of physical measurements or financial assets is reduced to a spanning tree. A vertex may be the estimated hub in every bootstrap tree while the available data still permit population trees in which it is peripheral. Degrees combine several edge decisions, and comparisons between correlations share observations and often a common vertex. An informative answer therefore needs both joint statistical calibration and joint tree restrictions.

This paper develops a checked route from edge-order inference to degree and rank statements covering every population-optimal tree. Its main statistical component averages betting e-values for raw Kendall contrasts across independent-of-data matchings. The finite-sample expectation bound covers ties directly and requires no moments; a size-dependent fraction gives consistency for fixed positive gaps. Conditional McNemar tests and dependence-safe p-value aggregation provide exact reference constructions. The computational component retains common-tree degree differences, adds cycle-path incompatibilities, verifies LP endpoint improvements by rational weak-duality certificates, and enforces the fixed sum of tree degrees. The combination can identify a degree or rule out a hub even when the whole optimal tree remains uncertain.

The cost of the exact construction is substantial: at continuous independence, a one-matching estimator has asymptotic variance 4.5 times that of the full U-statistic. We evaluate p-value aggregation over eight matchings and evidence aggregation over eight or 64 matchings, comparing these exact constructions with direct full-U-statistic Hoeffding tests and Gaussian multiplier

methods. Every calibration uses the same projections. Further ablations hold tests fixed while changing rank and path projection, and compare local with complete edge families. This separates gains from calibration, use of comparisons, and feasible-tree information.

The classical foundations are important to the contribution's scope. Interval robust spanning trees and possible or necessary optimality are studied by Yaman et al. (2001); Kasperski and Zieliński (2007); Fortin et al. (2009); Zaffalon and Hutter (2005) develop robust tree inference from dependence comparisons. The cut/cycle rules, greedy matroid rank identities and weak duality used here are established optimization facts. Our contribution is the explicit tie-aware matching-evidence construction and its consistency, the integration with verified degree-difference and path projection, and a matched evaluation of the resulting inferential trade-offs. Unlike interval optimization over supplied coordinate ranges, the input is a jointly calibrated relation between population edges, followed by projection to vertex quantities. Spanning-tree sensitivity analysis supplies related algorithmic context (Pettie, 2014).

Rank inference must retain population ties (Hall and Miller, 2009; Mogstad et al., 2024). Combinatorial inference for graphical models studies other graph properties under different structural models (Neykov et al., 2019). A correlation-optimal tree is an association summary. In financial networks, bootstrap edge reliability (Tumminello et al., 2007; Musciotto et al., 2018) and Polish market correlations (Gałązka, 2011) motivate the practical comparison. Edge or hub selection frequencies and simultaneous population degree statements are distinct statistical targets.

The primary measured-data example concerns Abalone morphometrics. It exhibits useful exact degree information and stronger full-sample bootstrap resolution under an explicitly stated cross-sectional sampling model. Polish returns supply a separate study of sensitivity to lookback and stock universe. A paired Pearson-bootstrap audit keeps implementation checks and calibration evidence separate. Together these examples expose both positive resolution and remaining limits of the procedure.

2 Population target and simultaneous orders

Let $G = (V, E)$ be a fixed connected graph with $p \geq 2$ vertices and m edges. For weights w, let

$$\mathcal{T}(w) = \arg \max_{T \text{ spanning tree}} \sum_{e \in T} w_e.$$

In the correlation application G is complete. A maximum-correlation tree has the same optimal-tree family as the minimum-distance tree based on $\sqrt{2(1 - w_e)}$: both are determined by the same edge ordering, including ties. Strictly increasing edge transformations preserve this family.

Pearson and Kendall targets. Gaussian and t_5 correlations have the same edge ordering as their Kendall weights, but this relation need not extend to a nonelliptical population. For a concrete construction, start with the $p = 12$ separated Gaussian star below and set $Y_i = \exp(\sigma_i X_i)$, where $\sigma_0 = 2.5$ and the other scales span $[0.3, 1.5]$. Monotonicity leaves every Kendall kernel unchanged. The normal exponential moment formula gives

$$\rho_{ij}^{Y} = \frac{\exp(\sigma_i \sigma_j \rho_{ij}^{X}) - 1}{\sqrt{(\exp(\sigma_i^2) - 1)(\exp(\sigma_j^2) - 1)}}.$$

The Kendall tree has vertex-zero degree 11, whereas the Pearson tree has vertex-zero degree 1. The supplied verifier checks both population trees and kernel invariance. Thus the weight definition is part of the inferential target.

Write $S_i = \{e : i \in e\}$ and $d_i(T) = |T \cap S_i|$. Descending ranks of vertex i, including every resolution of equal degrees, range from

$$r_i^-(T) = 1 + \sum_{j \neq i} \mathbf{1}\{d_j(T) > d_i(T)\}, \qquad r_i^+(T) = p - \sum_{j \neq i} \mathbf{1}\{d_j(T) < d_i(T)\}. \tag{1}$$

The inferential target contains these quantities for every $T \in \mathcal{T}(w_0)$. Under complete independence all edges have weight zero, every labelled tree is optimal, and the appropriate degree and rank targets are $[1, p-1]$ and $[1, p]$. Selecting one arbitrary tied optimum would artificially narrow this target.

A random strict relation $\widehat{\succ}$ is simultaneously valid if

$$\mathbb{P}(\mathcal{E}) \geq 1 - \alpha, \qquad \mathcal{E} = \{e \widehat{\succ} f \implies w_{0e} > w_{0f} \text{ for all } e, f\}. \tag{2}$$

We may take its transitive closure without an additional error budget. Every computational restriction below is conditional on this same event. The relation need not specify all pairwise edge comparisons, and its incompleteness is a source of conservative bounds.

3 An exact Kendall comparison construction

Suppose $X_1, \ldots, X_n$ are IID p-vectors. Coordinates may be dependent or discrete, and no moment assumption is imposed. For independent copies X, X', define

$$\tau_{ij} = \mathbb{E}[\operatorname{sgn}(X_i - X_i') \operatorname{sgn}(X_j - X_j')].$$

This is Kendall tau-a, not the tie-normalized tau-b. Attach independent uniform random variables U_{ti} to each observation and coordinate, and order $\widetilde{X}_{ti} = (X_{ti}, U_{ti})$ lexicographically. Randomization is independent of the observed sample.

For a predetermined pairing $(1,2), (3,4), \ldots$, let $N = \lfloor n/2 \rfloor$ and

$$S_{ki} = \operatorname{sgn}(\widetilde{X}_{2k-1,i} - \widetilde{X}_{2k,i}), \qquad H_{k,e} = S_{ki} S_{kj}, \quad e = \{i, j\}.$$

Here $H_{k,e} \in \{-1, 1\}$ and the vectors H_k are IID. For an ordered comparison (e, f) set

$$A_{ef} = \sum_k \mathbf{1}\{H_{k,e} = 1, H_{k,f} = -1\},$$
$$B_{ef} = \sum_k \mathbf{1}\{H_{k,e} = -1, H_{k,f} = 1\}, \qquad D_{ef} = A_{ef} + B_{ef}.$$

The one-sided exact p-value is

$$p_{ef} = \mathbb{P}\{\operatorname{Bin}(D_{ef}, 1/2) \geq A_{ef}\}, \tag{3}$$

with value one when $D_{ef} = 0$. This is the conditional McNemar test (McNemar, 1947), applied to paired edge kernels.

For a fixed family of unordered edge pairs $\mathcal{C}$, we test both directions, giving $K = 2|\mathcal{C}|$ hypotheses. The local family consists of distinct edges sharing a vertex and has $K = p(p-1)(p-2)$ on a complete graph. The complete family and its multiplicity cost are examined below. Apply Holm's step-down rule (Holm, 1979) at level α: ordered p-values $p_{(k)}$ are rejected until the first failure of $p_{(k)} \leq \alpha/(K - k + 1)$. Rejections define $e \widehat{\succ} f$; subsequently take the transitive closure.

Theorem 1 (Finite-sample order and hub coverage)**.** *For IID rows and the predetermined pairing above, the resulting relation satisfies* (2) *with $w_{0e} = \tau_e$, for every finite $n \geq 2$. The probability includes auxiliary tie randomization. Consequently, every degree, degree-difference and rank bound obtained by the valid outer projections in Sections 4–5 covers its target simultaneously with probability at least $1 - \alpha$, for all population-optimal trees and every resolution of degree ties.*

Proof. Conditional on X, X', if neither coordinate is tied, the augmented sign product equals the original sign product. If either is tied, the independent auxiliary sign for that coordinate has mean zero. Thus $\mathbb{E}H_{k,e} = \tau_e$, including when both coordinates tie. Put $\pi_+ = \mathbb{P}(H_e = 1, H_f = -1)$ and $\pi_- = \mathbb{P}(H_e = -1, H_f = 1)$. Then $\tau_e - \tau_f = 2(\pi_+ - \pi_-)$. Conditional on D_{ef}, A_{ef} is binomial with success probability $q = \pi_+/(\pi_+ + \pi_-)$ whenever the denominator is positive. Under $\tau_e \leq \tau_f$, $q \leq 1/2$, so (3) is superuniform by binomial stochastic monotonicity. The zero-denominator case gives $D_{ef} = 0$ almost surely. Holm's argument therefore controls the probability of any false strict order by α, regardless of dependence among comparisons. Transitive consequences remain true on the same event. Each subsequent outer projection retains all population-optimal trees on that event, proving the final assertion. □

The pairing cannot be selected after inspecting which arrangement gives favorable tests. A fixed pairing, or independent random pairing, is valid; trying several and retaining the best requires additional multiplicity control. This construction deliberately uses N independent pairs. The full-sample Kendall estimator uses all $n(n-1)/2$ pairs, whose kernels are dependent. Treating those kernels as independent would invalidate the binomial calculation. Applying the IID result to serially dependent financial returns or fitted residuals is also unjustified.

Proposition 2 (A sufficient discordance-dependent power bound)**.** *Consider $s \geq 1$ selected true comparisons from the tested family. Suppose each has gap $\Delta = \tau_e - \tau_f \geq \delta > 0$ and discordance probability $d = \mathbb{P}(H_e \neq H_f) \leq \bar{d}$. If*

$$N \geq \frac{64\bar{d}}{\delta^2} \max\left\{\log\frac{K}{\alpha}, \log\frac{2s}{\eta}\right\}, \qquad 0 < \eta < 1, \tag{4}$$

then all these comparisons are rejected with probability at least $1 - \eta$.

Proof. Let $Z_k = (H_{k,e} - H_{k,f})/2$. Its mean is $\Delta/2$, variance is at most $\bar{d}$, and its centered absolute value is at most two. Bernstein's inequality gives

$$\mathbb{P}\{A - B < N\delta/4\} \leq \exp\{-N\delta^2/(32\bar{d} + 16\delta/3)\} \leq \exp\{-N\delta^2/(64\bar{d})\},$$

where $\delta \leq 2\bar{d}$. A second Bernstein bound gives $\mathbb{P}(D > 2N\bar{d}) \leq \exp(-3N\bar{d}/8)$. Outside these events, the binomial Hoeffding bound implies

$$p_{ef} \leq \exp\{-(A-B)^2/(2D)\} \leq \exp\{-N\delta^2/(64\bar{d})\}.$$

Put $t = N\delta^2/(64\bar{d})$. Since $\delta^2 \leq 4\bar{d}^2$, both bad-event probabilities are at most e^{-t}. Condition (4) gives $e^{-t} \leq \alpha/K$ and $2se^{-t} \leq \eta$. A union bound over the s comparisons therefore gives the required probability, and every p-value at most α/K is rejected by Holm. □

The bound exhibits the role of discordance in simultaneous power. It is a sufficient condition; estimator efficiency and realized power are examined next.

3.1 Efficiency and repeated independent matchings

The disjoint-pair construction exchanges information for an exact conditional test. Randomized incomplete U-statistics offer related statistical/computational trade-offs (Chen and Kato, 2019). This cost is explicit at a continuous independence model. For adjacent edges $e = \{v, a\}$ and $f = \{v, b\}$, write $h = H_e - H_f$. Then $\mathrm{Var}(h) = 2$. After transforming the three margins to independent uniforms, its first-order projection is

$$g(U) = (2U_v - 1)\{(2U_a - 1) - (2U_b - 1)\}, \qquad \zeta_1 = \mathrm{Var}\{g(U)\} = 2/9.$$

The paired estimator has variance $2/N \sim 4/n$, while the full order-two U-statistic has asymptotic variance $4\zeta_1/n = 8/(9n)$. Their variance ratio is 4.5, corresponding to about 22% relative

efficiency. This calculation describes the continuous independence model; power also depends on the alternative, discordances and the simultaneous-testing family.

We therefore evaluate $R = 8$ matchings, the first predetermined and the others independent uniform permutations chosen independently of the measurements. Independent auxiliary tie randomization is used in each matching. Let $p_{ef}^{(r)}$ be its unadjusted one-sided p-value and define

$$\widetilde{p}_{ef} = \min\left\{1, \frac{2}{R}\sum_{r=1}^{R} p_{ef}^{(r)}\right\}. \tag{5}$$

For $R = 1$ we retain the original p-value. The factor two is the dependence-safe arithmetic-mean merging rule of Vovk and Wang (2020). Matchings share observations, so their p-values are dependent.

Proposition 3 (Finite-sample validity of matching aggregation)**.** *Under the sampling assumptions of Theorem 1, apply Holm to* (5) *over a fixed comparison family. Retain any subset of its rejections and then take transitive closure. The resulting strict orders, and every subsequent degree or rank projection, satisfy the same familywise guarantee* $1 - \alpha$*.*

Proof. Conditional on any independently chosen matching, each null p-value is superuniform by Theorem 1. It remains superuniform after averaging over the matching randomization. The arithmetic-mean rule is valid under arbitrary dependence, so every merged null p-value is superuniform. Holm controls any false rejection. Deleting rejections cannot increase this event, and the transitive consequences of true strict orders are true. □

For reproducible closure, the implementation retains only merged rejections consistent with the average paired edge scores. This deletion preserves the guarantee. It does not introduce comparisons that failed Holm. Aggregating p-values reduces dependence on an individual matching, but its power must be measured: the factor two and p-value averaging can be conservative. For comparison, averaging R independently randomized matching *estimators* has variance $\mathrm{Var}(U) + (\mathrm{Var}(V) - \mathrm{Var}(U))/R$, by conditional expectation over matchings. At the independence model its asymptotic variance relative to the full U-statistic is $1 + 3.5/R$. This estimator calculation alone does not establish an equivalent efficiency for (5).

3.2 Exact evidence aggregation with deterministic tie handling

P-value averaging can be conservative at a large comparison family. We therefore also aggregate nonnegative evidence with null expectation at most one, an e-value (Vovk and Wang, 2021). For matching r, use the original tau-a kernels $K_{r,k,e} \in \{-1, 0, 1\}$, retaining zero at a tied coordinate, and set $Z_{r,k,ef} = (K_{r,k,e} - K_{r,k,f})/2 \in [-1, 1]$. For a predetermined fraction $0 \le \lambda < 1$, define

$$E_{r,\lambda,ef} = \prod_{k=1}^{N}(1 + \lambda Z_{r,k,ef}).$$

The evaluated multiset of seven fractions is

$$\Lambda_N = \{1/32, 1/16, 1/8, 1/4, 1/2, 3/4, \min(1/2, N^{-1/2})\}.$$

Its last element is determined by sample size and ensures consistency even for gaps too small for the fixed fractions. We compare $R = 8$ and $R = 64$ matchings, with the former nested in the latter, and put

$$\overline{E}_{ef} = \frac{1}{7R}\sum_{r=1}^{R}\sum_{\lambda\in\Lambda_N} E_{r,\lambda,ef}, \qquad p_{ef}^{E} = \min\{1, 1/\overline{E}_{ef}\}. \tag{6}$$

The averaging rule is classical e-value merging. Its use here combines exact Kendall contrast evidence across shared-observation matchings. The statistic handles ties directly, without auxiliary sign randomization. Matching randomization uses the same stream as the p-value comparator, with unused tie draws retained solely to align the matching sequences.

Proposition 4 (Finite-sample evidence validity and fixed-family consistency)**.** *Under IID rows, Holm applied to* (6) *controls false strict orders at level α, including discrete margins. Deleting rejections inconsistent with the full-sample Kendall scores and taking transitive closure preserves this control. For fixed R and a fixed comparison family, each comparison with $\tau_e > \tau_f$ is rejected with probability tending to one as $n \to \infty$.*

Proof. For a fixed independently selected matching, the contrasts Z_k are IID with $\mathbb{E}Z_k = (\tau_e - \tau_f)/2$. Under the null this mean is nonpositive, so nonnegativity and independence give

$$\mathbb{E}E_{r,\lambda,ef} = \{1 + \lambda \mathbb{E}Z_k\}^N \leq 1.$$

Averaging over fractions and matchings preserves the expectation bound regardless of their dependence. Markov's inequality makes the reciprocal p-value superuniform. Holm, deletion and transitive closure give the common event used by all projections.

For a fixed positive gap Δ, take the first matching and $\lambda_N = N^{-1/2}$ for $N \geq 4$. Since $\log(1+u) \geq u - u^2$ for $|u| \leq 1/2$,

$$\log E_{1,\lambda_N,ef} \geq N^{-1/2} \sum_k Z_k - N^{-1} \sum_k Z_k^2 \geq \sqrt{N}\{\Delta/2 + o_{\mathbb{P}}(1)\} - 1 \;\longrightarrow\; \infty.$$

Thus $\overline{E}_{ef} \geq E_{1,\lambda_N,ef}/(7R)$ diverges in probability, and $p^E_{ef} \to 0$. The full-sample Kendall difference converges to Δ, so the consistency filter retains the direction with probability tending to one. Holm's threshold is at least $\alpha/K > 0$ for a fixed family, proving the assertion. □

The result establishes validity and consistency; relative power varies by design and is measured below. The simulations compare both evidence averages with the same eight-matching p-value pool and with full-U-statistic methods. For continuous margins and local comparisons, $H_{\{v,a\}}H_{\{v,b\}} = S_a S_b$: one $p \times p$ sign Gram matrix supplies all discordance counts in $O(Np^2 + K)$ work per matching. Tied raw kernels use exact five-category contrast counts. Logarithmic accumulation prevents numerical overflow of products, while independent direct-kernel checks verify the counts and evidence calculations.

3.3 A finite-sample comparator using all observation pairs

Define $k_e(X_s, X_t) = \operatorname{sgn}(X_{si} - X_{ti}) \operatorname{sgn}(X_{sj} - X_{tj})$ and

$$\widehat{\tau}_e = \binom{n}{2}^{-1} \sum_{s<t} k_e(X_s, X_t), \qquad \widehat{\Delta}_{ef} = \widehat{\tau}_e - \widehat{\tau}_f.$$

The difference kernel lies in $[-2, 2]$, including ties. Hoeffding's order-two U-statistic inequality (Hoeffding, 1963) gives

$$\mathbb{P}\{\widehat{\Delta}_{ef} - \Delta_{ef} \geq t\} \leq \exp(-Nt^2/8). \tag{7}$$

Consequently $p^U_{ef} = \exp\{-N(\max\{\widehat{\Delta}_{ef}, 0\})^2/8\}$ is superuniform under $\Delta_{ef} \leq 0$. Holm and the same projections therefore give finite-sample simultaneous inference. To see (7) directly, average the disjoint-pair mean over a uniform random matching conditional on the sample. This conditional average is $\widehat{\Delta}_{ef}$. Jensen's inequality bounds its exponential moment by that of a mean of N independent kernels of range four; optimizing the usual bounded-variable exponential bound gives (7). Thus dependence between the $\binom{n}{2}$ kernels is respected.

This comparator isolates direct testing from rectangularization while retaining a finite-sample guarantee. Its bounded-range calibration can remain conservative even though its estimator uses all pairs.

3.4 Full U-statistics with Gaussian multiplier calibration

For a computationally stronger comparator, estimate the first-order influence by

$$\widehat{\psi}_{te} = 2\left\{\frac{1}{n-1}\sum_{s\neq t} k_e(X_s, X_t) - \widehat{\tau}_e\right\}.$$

Given independent standard normal multipliers $\xi_t^{(b)}$, put $Z_e^{*(b)} = n^{-1}\sum_t \xi_t^{(b)}\widehat{\psi}_{te}$ and

$$s_{ef}^2 = n^{-2}\sum_t (\widehat{\psi}_{te} - \widehat{\psi}_{tf})^2, \qquad Q_b = \max_{\{e,f\}\in\mathcal{C}} \frac{|Z_e^{*(b)} - Z_f^{*(b)}|}{s_{ef}}.$$

The empirical $1-\alpha$ quantile q gives intervals $\widehat{\Delta}_{ef} \pm q s_{ef}$. Intervals excluding zero supply strict orders. This is an asymptotic IID procedure for nondegenerate projections, grounded in Gaussian and multiplier approximation for high-dimensional U-statistics (Chen, 2018; Chernozhukov et al., 2013). Theorem 1 applies to the exact methods above; multiplier calibration has its own approximation assumptions. Numerically degenerate contrasts are omitted, and their counts are reported.

The matched multiplier rectangular comparator uses the same scores, influences and multiplier draws, calibrates the maximum standardized error across individual edges, and converts its edge intervals into strict orders over the same family $\mathcal{C}$. Both comparators then use identical tree projections. This comparison measures the effect of calibrating differences rather than separate edge weights.

Tie-safe forward and backward Fenwick sweeps compute every rowwise kernel mean in $O(p^2 n \log n)$ time. Contrast maxima are streamed in C++ without allocating a $B \times |\mathcal{C}|$ matrix. These implementations retain the full tau-a statistic and its shared-edge covariance.

3.5 Local and complete families

Every fixed comparison family can be used in the exact constructions. We compare all pairs sharing a vertex with all pairs of distinct edges. Their ordered hypothesis counts on a complete graph are $K_{\text{local}} = p(p-1)(p-2)$ and $K_{\text{all}} = m(m-1)$, with ratio $(p+1)/4$ for $p \geq 3$. At $p = 52$ these are 132,600 and 1,756,950.

Complete comparisons can add nonlocal information relevant to cycle-path constraints, while also increasing the multiplicity penalty and computation. Transitive closure of local orders recovers some nonlocal comparisons, but need not recover all of them. The family ablation measures the resulting trade-off at $p = 12$; both families are also evaluated on every cross-sectional application. Local testing remains an explicit computational choice for the larger simulations and the Polish sensitivity study.

4 Projecting orders into joint degree information

4.1 Classical screening and exact extrema on an outer minor

For an edge $e = \{u, v\}$, exclude e if a u–v path consists entirely of edges certified above e. The cycle property rules it out of every population-optimal tree. Force e if u, v become disconnected after deleting e and all edges certified below e. The resulting cut has e strictly heavier than every alternative, so every optimum contains it. These sufficient screening rules define the outer family used in the subsequent projections.

Let A be the allowed edges and F the forced edges. On $\mathcal{E}$, F is a forest. Contract F, delete excluded edges, and retain the resulting graphic matroid M with rank $k = p - 1 - |F|$ and rank function r. Parallel edges and loops are permitted after contraction. Every population-optimal

tree is $F \cup B$ for some base B of M, but many bases can fail to be optimal for any compatible weights. All exactness claims below refer to this outer base family.

Write $a_i = |F \cap S_i|$ and $I_i = S_i \setminus F$ restricted to the residual ground set. The ordinary greedy rank identities give

$$\underline{d}_i = a_i + k - r(E_M \setminus I_i), \qquad \overline{d}_i = a_i + r(I_i). \tag{8}$$

For vertices i, j, assign residual coefficients $c_e = \mathbf{1}\{e \in S_i\} - \mathbf{1}\{e \in S_j\}$, and let P_+, P_0, P_- denote their level sets.

Proposition 5 (Degree-difference extrema on the minor)**.** *The exact maximum and minimum of $d_i - d_j$ over the outer base family are*

$$U_{ij} = a_i - a_j + r(P_+) + r(P_+ \cup P_0) - k, \tag{9}$$

$$L_{ij} = a_i - a_j + k - r(P_-) - r(P_- \cup P_0). \tag{10}$$

Proof. For a base B of size k, $\sum_{e \in B} c_e = |B \cap P_+| + |B \cap (P_+ \cup P_0)| - k$. Each intersection is bounded by the corresponding rank. A greedy base processing P_+ first, then P_0, then P_- attains both bounds simultaneously, by the matroid augmentation property. This proves (9). Reverse signs for (10). The same argument with coefficients zero and one gives (8). □

These classical matroid identities give an explicit inferential calculation: $L_{ij} > 0$ certifies $d_i > d_j$ for every retained tree. Directly optimizing a difference retains a common feasible tree, whereas subtracting a marginal maximum from a marginal minimum can use incompatible trees.

4.2 Cycle-path incompatibilities

Individual screening can leave many jointly impossible edge configurations. Suppose P is a simple path between the endpoints of e, and $e \widehat{\succ} f$ for some $f \in P$. Every population-optimal tree satisfies

$$\sum_{g \in P} x_g \leq |P| - 1, \tag{11}$$

where x_g is its edge indicator. Indeed, a tree containing all of P cannot also contain e; replacing f by e would then strictly increase its weight. Notice that the condition needs one lighter edge on the path. It is stronger than excluding only an edge that is lighter than an entire alternative path.

We use every such path of length two and additional paths from eight candidate spanning trees. Candidate priorities are generated with a fixed seed, with forced edges processed first and excluded edges omitted. For each candidate and each non-tree edge, its unique path is checked against the same certified relation. Path selection may depend on the relation without a new statistical correction: every inserted inequality is valid on $\mathcal{E}$. Eight is a computational design choice, not a claim that these paths exhaust the useful constraints.

The linear relaxation has $0 \leq x_e \leq 1$, $\sum_e x_e = p - 1$, $\sum_{e \in S_i} x_e \geq 1$, fixed forced/excluded coordinates, and the selected inequalities (11). To impose connectivity, add directed nonnegative flows, root outflow minus inflow $p - 1$, and outflow minus inflow -1 at every other vertex. For an undirected edge $e = \{u, v\}$ impose

$$f_{uv} + f_{vu} \leq (p - 1)x_e.$$

Every tree admits these flows by routing one unit from the root to each other vertex. Fractional solutions may remain, so this is an outer relaxation, not the convex hull of compatible optimal trees.

In the new small-graph simulations and cross-sectional applications we minimize and maximize every degree and every ordered degree difference. The larger Polish sensitivity analysis optimizes degree differences between the observed highest-degree anchor and every other vertex, retaining minor bounds for all remaining pairs. Anchor choice uses the data but all projected statements use the same event $\mathcal{E}$.

4.3 Rational certificates for numerical optimization

A floating-point optimum should not be rounded into an inferential statement without a valid bound. Write the relaxation as $Ax \le b$, $Ex = h$, $\ell \le x \le u$, incorporating both edge and flow variables. Its coefficients and bounds are integers. For an integer objective c, take any rational multipliers $y \le 0$ and unrestricted z, and put $v = c - A^\top y - E^\top z$.

Proposition 6 (Verifiable objective bound)**.** *For every feasible x,*

$$c^\top x \ge b^\top y + h^\top z + \sum_j \min\{v_j \ell_j, v_j u_j\} =: q. \tag{12}$$

If the target objective is integer, $\lceil q \rceil$ is a valid lower bound. Applying the formula to $-c$ gives upper bound $\lfloor -q \rfloor$.

Proof. Expand $c^\top x = y^\top Ax + z^\top Ex + v^\top x$. The first term is at least $y^\top b$ because $y \le 0$; the second equals $z^\top h$; coordinatewise minimization over the box bounds the last term. Integer rounding preserves the assertion. □

The implementation takes inequality multipliers from HiGHS, clips them to be nonpositive and rounds downward to multiples of 2^{-20}. Equality multipliers are rounded to that denominator. The residual and bound are then computed with checked integer arithmetic. Empirical sparse certificates contain the numerator, denominator and multipliers for every improvement. Simulation improvements are verified during each replication, with verification counts and replication seeds archived. An independent verifier reconstructs the constraints and recomputes the bound with Python arbitrary-precision integers. Validity does not require that the numerical solution satisfy KKT equations exactly, or even that its proposed multipliers be optimal.

5 Rank projection with degree conservation

Let integer degree bounds be $L_i \le d_i \le U_i$, and impose

$$\sum_i d_i = 2p - 2 =: D_0, \qquad d_i \ge 1. \tag{13}$$

We compute sharp rank endpoints over this integer box intersected with (13). For vertex i, enumerate feasible $s = d_i \in [L_i, U_i]$, so that $\sum_{j \ne i} L_j \le D_0 - s \le \sum_{j \ne i} U_j$.

For the worst rank, all vertices with $L_j \ge s$ are already tied with or above i. Each optional vertex with $L_j < s \le U_j$ costs $s - L_j$ additional degree units to reach that threshold. Starting from all lower bounds, select the cheapest such costs within budget $D_0 - s - \sum_{j \ne i} L_j$. One plus the mandatory and selected counts is the largest possible rank for that s.

For the best rank, vertices with $L_j > s$ necessarily outrank i. Without raising any additional vertex above s, the largest possible total for the other degrees is

$$C(s) = \sum_{j : L_j > s} U_j + \sum_{j : L_j \le s} \min(U_j, s).$$

If $D_0 - s > C(s)$, select the fewest additional vertices whose capacities $U_j - s > 0$ cover the shortfall, taking largest capacities first. Add their count and the mandatory count to one. Minimize this best rank and maximize the worst rank over feasible s.

Proposition 7 (Sharpness within the degree relaxation)**.** *The threshold and capacity calculations above attain the best and worst ranks over all integer vectors satisfying the degree bounds and* (13)*. On the complete graph every positive integer vector with this total is the degree sequence of a labelled tree. Sharpness need not hold for the smaller family of order-compatible or minor-compatible trees.*

Proof. To make k optional competitors reach s, their required increases have minimum total equal to the sum of the k smallest costs. The budget criterion is therefore necessary and sufficient; remaining degree units can be allocated within the original upper bounds. If this forced another optional competitor to reach s, that competitor would already have been affordable, contradicting maximality. For the best rank, activating a vertex above s supplies at most $U_j - s$ additional units. The largest-capacity rule minimizes the number required. Feasibility of s guarantees enough total capacity, and integer intervals allow filling the remaining total without gaps. Finally, a Prüfer sequence containing vertex i exactly $d_i - 1$ times realizes any positive vector summing to $2p - 2$. □

Intersect these rank intervals with direct comparison bounds

$$1 + \#\{j : L_{ji} > 0\} \ \le r_i \le \ p - \#\{j : L_{ij} > 0\}. \tag{14}$$

A certified unique hub has upper rank one. A vertex remains a possible hub whenever its lower rank is one. Estimated degrees are reported separately from these tie-inclusive inferential ranges. The proof of Theorem 1 now follows through every computational stage.

6 A positive-definite example of lost joint information

Consider eight vertices indexed $0, \dots, 7$. In the block $\{0, \dots, 5\}$ let hub correlations be $\rho_{0j} = 0.06 + t$ for $1 \le j \le 5$, and leaf correlations $\rho_{ij} = 0.04 + t$ for $1 \le i < j \le 5$. Set $\rho_{06} = 0.10$, $\rho_{67} = 0.15$, all other cross-block correlations to zero, and diagonal entries to one, with $t \in [-0.025, 0.025]$.

Every matrix is positive definite: it is symmetric, has positive diagonal and is strictly diagonally dominant; the largest absolute off-diagonal row sum is 0.525. Every within-block spoke exceeds every within-block leaf edge by 0.02. The unique optimal tree throughout the curve consists of five spokes, the bridge 06 and edge 67. Its degree vector is $(6, 1, 1, 1, 1, 1, 2, 1)$, so vertex zero is uniquely first.

Now transform correlations by Fisher's atanh and subtract the mean across all 28 edges at each t. Form the smallest coordinatewise rectangular hull of this centered curve. Global centering removes a common shift across all edges but not a shift restricted to one block. Exact rectangular projection yields degree ranges

$$[2, 6],\ [1, 5] \text{ (each of five leaves)},\ [2, 2],\ [1, 1].$$

The marginal rank construction gives vertex zero $[1, 7]$. In contrast, the true local orders along the coupled curve force all seven tree edges and rank $[1, 1]$. For a changing edge, the derivative of its centered Fisher weight is bounded below by $1 - 15/[28(1 - 0.085^2)] > 0$; for a constant edge it is negative. Thus its coordinate extrema occur at the two endpoints. An executable verification is supplied with the code. This is a deterministic comparison of uncertainty-set shapes, not a claim that this curve is an estimated confidence region. It shows why rectangularization can discard relevant information even when the candidate matrices are positive definite and global centering has already been used.

7 Matched simulation comparisons

7.1 Models, targets and calibration

We evaluate eight calibrations on identical data: one-matching exact McNemar–Holm, eight-matching p-value aggregation, evidence aggregation over eight or 64 matchings, direct full-U-statistic Hoeffding–Holm, the full-sample rectangular Hoeffding comparator, Gaussian multiplier contrasts, and Gaussian multiplier edge rectangles. The rectangular methods produce orders

over the same fixed family as their direct comparators, and every method passes through the same projections.

For the separated Gaussian model, $X_i = a_i Z + \sqrt{1 - a_i^2}\epsilon_i$, with independent standard normals, $a_0 = 0.99$ and the other loadings equally spaced over $[0.45, 0.55]$. The near-tie model uses $a_0 = 0.8 + 0.5/\sqrt{n}$, $a_1 = 0.8$, with other loadings in $[0.45, 0.55]$. The t_5 model divides each Gaussian row by an independent $\sqrt{\chi_5^2/5}$. These models have a unique population star centred at zero; their population Kendall weights equal $2\arcsin(\rho)/\pi$.

Gaussian independence and IID uniform coordinates on $\{-1, 0, 1\}$ evaluate null comparisons and exact ties. Every labelled tree is optimal in these null models, so coverage must include degree ranges $[1, p-1]$ and rank ranges $[1, p]$, rather than the degrees of one tie-broken tree. Two additional Gaussian tree models avoid restricting the comparison to a uniquely dominant star. The chain has parent correlation 0.75; the two-hub model has bridge correlation 0.5 and leaf-parent correlation 0.7, with five leaves attached to each hub at $p = 12$. All population tree edges are unique optima, but equal vertex degrees retain their full tied rank ranges.

The 24 cells comprise 7,800 independent datasets. At $p = 12$, the three star designs use $n \in \{252, 1008, 4096\}$ and 200 datasets per cell. At $p = 52$, they use $n \in \{252, 4096\}$ and 100 per cell. Gaussian null cells at $p = 12$ use all three sizes, and at $p = 52$ use $n = 1008$; the discrete null uses $p = 12, n = 1008$. Every null cell has 1,000 datasets. Chain and two-hub cells use $p = 12, n \in \{1008, 4096\}$ and 100 datasets per cell. All tests use $\alpha = 0.05$, and multiplier calibration uses 999 draws. Seeds, complete cell definitions and individual outcomes are archived.

We report false-order FWER, simultaneous degree and rank coverage, interval widths, forced edges, possible hubs and unique-hub declarations. Wilson intervals quantify Monte Carlo uncertainty. With 1,000 repetitions, the Monte Carlo standard error at probability 0.05 is about 0.007. A single empirical rejection proportion is not interpreted as a proof or disproof of a coverage theorem.

7.2 Test, projection and family ablations

For every test we compare marginal rank projection, direct degree-difference projection, degree conservation, and their intersection, using the same screened minor. This isolates computational changes from statistical calibration. For the first 100 datasets in every $p = 12$ cell, we also add the cycle-path LP and optimize all ordered degree differences, rather than only anchor differences. Improvements are checked independently with Python arbitrary-precision integers. Empty order relations and a forced single tree have known sharp endpoints, so redundant LP solves are skipped. Path-stage estimates therefore have 100 repetitions per cell and are compared with the corresponding paired subset of the other stages.

The complete-edge comparison family is evaluated on the same first 100 non-null datasets in each $p = 12$ cell. All four $p = 12$ null cells extend both families to 1,000 datasets. Comparisons of families use matched subsets; differences in reported means do not arise from different simulated samples.

7.3 A frequency reference from practice

Bootstrap edge reliability is widely used in correlation networks (Tumminello et al., 2007; Musciotto et al., 2018). On the same $p = 12$ separated-star, near-tie and Gaussian-null samples with $n = 1008$ and 4096, we compute 499 ordinary Pearson row-bootstrap trees. Pearson and Kendall have the same population optimal-tree ordering in these Gaussian designs. We record edge selection frequencies and the frequency with which the observed anchor is the unique highest-degree vertex. A threshold of 0.95 is evaluated as a practical declaration rule, including its false hub and false forced-edge rates. It is a frequency heuristic, not a simultaneous confidence procedure. This reference uses all 200 non-null and all 1,000 null datasets in the six selected cells.

7.4 Calibration and power results

Table 1 reports null FWER for the common minor-plus-joint-rank projection. Exact methods have their finite-sample guarantee; the multiplier methods are evaluated empirically. All cellwise Wilson intervals, simultaneous coverage outcomes and replication records accompany the article. The largest one-matching null estimate is 0.035, with an individual Wilson interval [0.025,0.048]

Table 1: False-order FWER at nominal 0.05; 1,000 datasets in each null cell. P: one-matching exact; M8: merged p-values; E8/E64: evidence aggregation over 8/64 matchings; M-C: multiplier contrasts; M-B: multiplier rectangle. Both Hoeffding controls have zero false orders in these cells. All methods use the local family.

Model, p, n	P	M8	E8	E64	M-C	M-B
Gaussian null, 12, 252	0.024	0.000	0.000	0.001	0.072	0.002
Gaussian null, 12, 1008	0.029	0.000	0.000	0.000	0.062	0.003
Gaussian null, 12, 4096	0.032	0.000	0.001	0.000	0.042	0.001
Discrete null, 12, 1008	0.029	0.000	0.001	0.000	0.050	0.001
Gaussian null, 52, 1008	0.035	0.000	0.000	0.000	0.077	0.000

(Gaussian null, $p = 52$, $n = 1008$). This is a cellwise Monte Carlo summary; simultaneous calibration is established by the exact-test argument. Power is interpreted jointly with FWER.

Table 2: Probability of declaring the true unique hub using the same minor and joint-rank projection. Local comparisons; 200 repetitions for the displayed $p = 12$ star cells and 100 for $p = 52$. Both Hoeffding controls declare no unique hub in these star cells; full results are archived. Path projection is evaluated separately on paired subsets.

Model, p, n	P	M8	E8	E64	M-C	M-B
Gaussian star, 12, 1008	0.000	0.000	0.000	0.000	1.000	0.900
Gaussian star, 12, 4096	1.000	0.990	1.000	1.000	1.000	1.000
t_5 star, 12, 1008	0.000	0.000	0.000	0.000	1.000	0.195
t_5 star, 12, 4096	1.000	1.000	1.000	1.000	1.000	1.000
Gaussian star, 52, 4096	0.670	0.000	1.000	1.000	1.000	1.000
t_5 star, 52, 4096	0.720	0.000	1.000	1.000	1.000	1.000
Near-tie star, 12, 4096	0.000	0.000	0.000	0.000	0.000	0.000
Near-tie star, 52, 4096	0.000	0.000	0.000	0.000	0.000	0.000

The multiplier contrast procedure exceeds nominal error in some IID null cells; higher declaration rates in those regimes do not establish valid superiority. The exact evidence methods provide a comparison whose error guarantee does not rely on the Gaussian approximation. Figure 1 compares the full sample-size grids. The merged rule is assessed as an exact alternative, rather than assumed to recover U-statistic efficiency. The bounded-range U-statistic comparator shows whether using all pairs alone is sufficient; the matched multiplier rectangle and contrast procedures isolate the effect of shared-edge covariance and direct calibration.

7.5 Projection ablation

Table 3 locates the path-projection gain mainly in degree precision. For one-matching inference, Gaussian-star mean degree width decreases from 8.94 to 8.40; in the chain it decreases from 8.00 to 4.58. The corresponding E64 chain widths decrease from 6.48 to 3.97. Most exact rank widths remain approximately 11, the full-range width at $p = 12$. In the near-tie multiplier example, the path stage decreases mean rank width only from 10.90 to 10.81. Thus these $n = 1008$ simulations demonstrate useful degree restrictions but little additional rank resolution: tighter degree bounds need not establish a positive degree difference between two vertices.

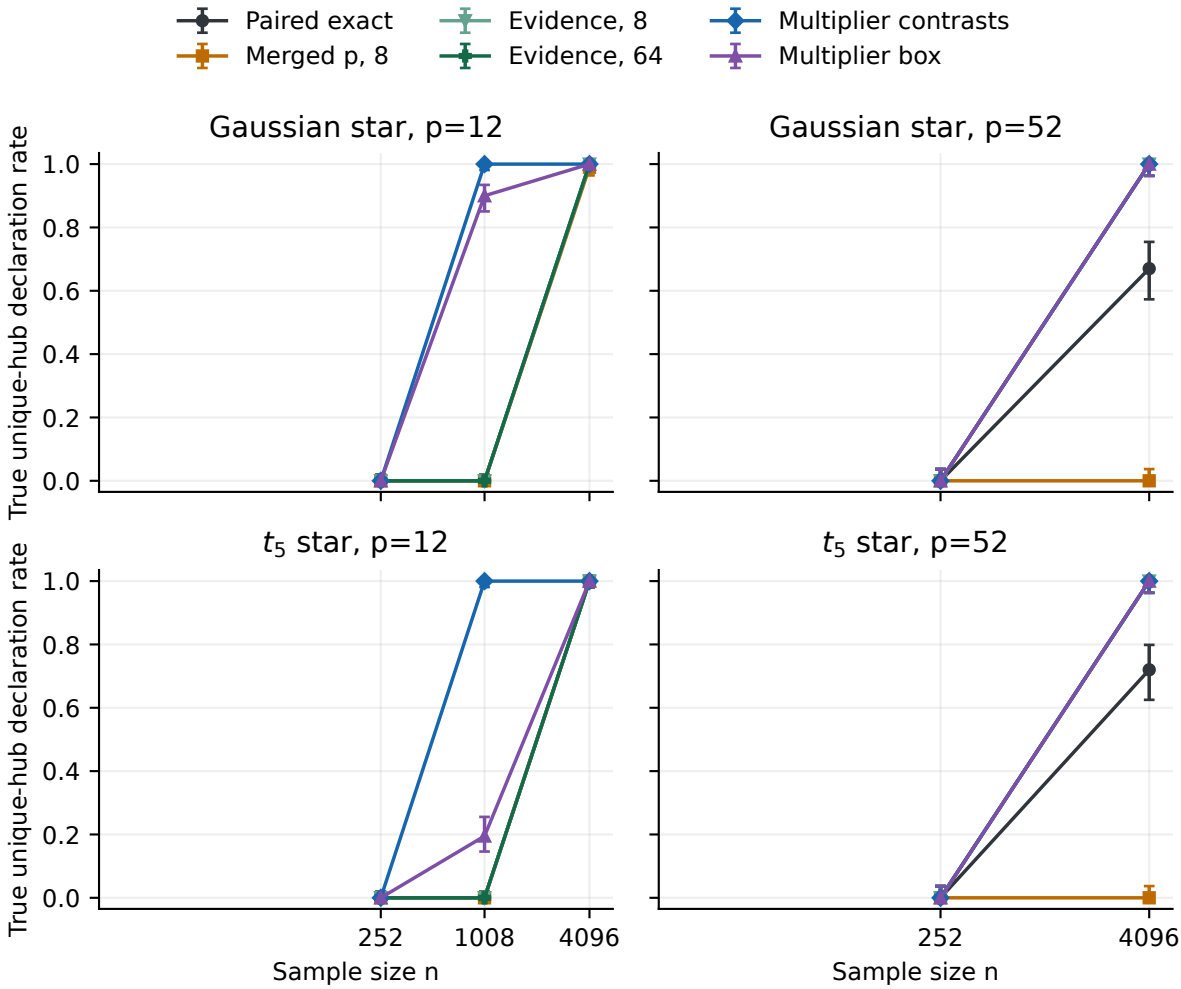


Figure 1: Unique-hub declarations under the common joint projection. Curves use matched data; intervals are individual 95% Wilson Monte Carlo intervals. All specified sample sizes and both separated distributions are shown.

Table 3: Paired projection ablation at $p = 12, n = 1008$, using the same first 100 datasets per cell. Degree widths compare the minor with path LP. Rank widths compare marginal, joint pair/conservation and path projection. Complete results for all eight tests and sizes are archived. Width is upper minus lower endpoint.

Model	Test	W_d minor	W_d path	W_r marg.	W_r joint	W_r path
Gaussian star	P	8.94	8.40	11.00	11.00	11.00
Gaussian star	E64	9.02	8.33	11.00	11.00	11.00
Gaussian star	M-C	0.00	0.00	9.17	9.17	9.17
Near-tie star	P	9.92	9.45	11.00	11.00	11.00
Near-tie star	E64	9.99	9.51	11.00	11.00	11.00
Near-tie star	M-C	3.93	3.54	11.00	10.90	10.81
Chain	P	8.00	4.58	11.00	11.00	11.00
Chain	E64	6.48	3.97	11.00	10.99	10.99
Chain	M-C	0.00	0.00	7.67	7.67	7.67
Two hubs	P	8.43	6.66	11.00	10.96	10.96
Two hubs	E64	5.82	5.11	10.98	10.54	10.54
Two hubs	M-C	0.58	0.58	7.67	7.67	7.67

Population degree ties also impose an irreducible rank width. Even with the exact Gaussian-star tree known, its eleven leaves occupy ranks 2–12 under the tie-inclusive target, giving mean rank width $110/12 = 9.17$. For the chain and two-hub trees, the corresponding mean is $92/12 = 7.67$. The displayed multiplier results at these widths therefore retain the required tied ranks. Degree-conservation and direct-pair rank outputs are archived separately from the path stage.

Across all sample sizes and methods in the $p = 12$ simulation path stage, the local family accounts for 950,820 LP objectives and 375,946 independently verified improving rational certificates. The complete family adds 913,068 objectives and 273,901 certificates. The totals across both families are therefore 1,863,888 LP objectives and 649,847 certificates. These totals include all evaluated path-stage cells, rather than only the subset displayed in Table 3.

7.6 Comparison-family ablation

Table 4: Matched local/complete family ablation at $p = 12, n = 4096$, 100 datasets per cell. Closed-order counts use the same $m(m-1)$ potential ordered pairs in either family. More tested comparisons also incur a larger multiplicity penalty.

Model	Test	Orders L	Orders A	W_d L	W_d A	W_r L	W_r A
Gaussian star	P	113.0	530.7	0.14	0.25	9.17	9.17
Gaussian star	E64	110.8	579.0	0.00	0.00	9.17	9.17
Gaussian star	M-C	214.4	643.0	0.00	0.00	9.17	9.17
t_5 star	P	112.0	527.8	0.17	0.31	9.17	9.17
t_5 star	E64	110.5	573.7	0.00	0.00	9.17	9.17
t_5 star	M-C	193.3	629.0	0.00	0.00	9.17	9.17
Near-tie star	P	137.3	331.6	6.40	7.20	11.00	11.00
Near-tie star	E64	170.8	393.1	5.53	6.54	10.99	11.00
Near-tie star	M-C	331.1	969.4	1.66	1.67	10.16	10.17
Chain	P	850.2	1338.6	0.14	0.00	7.98	7.67
Chain	E64	923.0	1396.0	0.00	0.00	7.67	7.67
Chain	M-C	1374.2	1766.2	0.00	0.00	7.67	7.67
Two hubs	P	624.1	1117.3	0.99	0.38	7.67	7.67
Two hubs	E64	659.5	1157.1	0.97	0.02	7.67	7.67
Two hubs	M-C	725.3	1538.1	0.00	0.00	7.67	7.67

Table 4 shows a model-dependent trade-off. For one-matching inference in the Gaussian star, complete comparisons increase mean degree width from 0.14 to 0.25 despite increasing the closed-order count from 113.0 to 530.7. In the near-tie E64 example, width increases from 5.53 to 6.54. Conversely, complete comparisons improve the one-matching chain width from 0.14 to 0.00 and the two-hub width from 0.99 to 0.38; for E64 in the two-hub model they reduce it from 0.97 to 0.02.

The procedures recalibrate their tested families, so their rejected-order relations need not be nested. At $p = 12$, the local and complete families contain 1,320 and 4,290 ordered hypotheses. For the Holm methods, the first threshold decreases from $0.05/1320$ to $0.05/4290$; multiplier contrasts also recalibrate a larger maximum. This can remove weak local comparisons important to screening incident hub edges, while adding many nonlocal orders. The number of certified orders alone therefore does not determine degree resolution. Holding one valid relation fixed and adding further valid comparisons can only tighten exact bounds over the resulting outer family; changing the calibrated relation is a different comparison. Complete testing is useful when its additional tree restrictions outweigh its multiplicity cost, as in the chain and two-hub examples.

7.7 Frequency-reference results

Table 5: Practical row-bootstrap frequency rule on matched Gaussian samples, $p = 12$, 499 draws. Hub declarations require unique-hub frequency at least 0.95; edge declarations require selection frequency at least 0.95. False edge declarations are familywise events. Frequencies have no general confidence guarantee.

Model	n	R	True hub	False hub	False edge
Gaussian star	1008	200	1.000	0.000	0.000
Gaussian star	4096	200	1.000	0.000	0.000
Gaussian null	1008	1000	–	0.000	0.043
Gaussian null	4096	1000	–	0.000	0.049
Near-tie star	1008	200	0.020	0.000	0.220
Near-tie star	4096	200	0.060	0.000	0.225

The frequency reference targets estimated-tree stability. Under the Gaussian null every edge can occur in an optimal population tree, but none is necessary and no vertex is a population unique hub. A high-frequency edge declaration is therefore counted as false necessity in that setting. This definition avoids interpreting frequent selection as a simultaneous population certificate.

8 Paired audit of Pearson-bootstrap calibration

The original diagnostic uses fixed bootstrap standard deviations for local Fisher-z Pearson differences. It showed substantial undercoverage in some IID and serially dependent cells. To identify which features matter, the new audit holds datasets fixed and varies bootstrap mechanism, studentizer and resample count.

Three IID Gaussian independence cells, $(p, n) = (12, 252), (52, 252), (52, 1008)$, each contain 1,000 datasets. We compare the original row bootstrap with 499 and 1,999 resamples, an IID row bootstrap with empirical influence standard deviations and 1,999 resamples, and Gaussian influence multipliers with 1,999 draws. The 499 row draws are nested in the 1,999 draws. For Pearson correlation, standardized observations z_{ti} give influence

$$\phi_{t,ij} = \frac{z_{ti}z_{tj} - \widehat{\rho}_{ij}(z_{ti}^2 + z_{tj}^2)/2}{1 - \widehat{\rho}_{ij}^2}.$$

We centre these influences across rows and use $n^{-2}\sum_t(\phi_{te} - \phi_{tf})^2$ for a contrast variance. Every calibration uses the same local family, Fisher-z estimand and maximum statistic. Under independence, every population contrast is zero, making input undercoverage identical to any false strict order. We report paired differences between procedures, not just isolated estimates.

Direct index resampling checks the count-weighted covariance implementation. Independent calculations check influences, studentizers and maxima. This audit evaluates particular procedures at these sample sizes; it separates an implementation check from a conclusion about bootstrap calibration. The earlier 24-cell diagnostic, including serial dependence, is retained in the reproducibility materials.

At $p = 52, n = 1008$, the paired FWER difference (499 minus 1,999 row draws with the original studentizer) is 0.021, with Monte Carlo interval [0.007,0.035] based on paired replication differences. The studentizer and mechanism comparisons in Table 6 provide separate diagnostics. The verified resampling identities support implementation correctness; the calibration results concern the particular finite-sample procedures.

Table 6: Paired Gaussian IID null audit. Each procedure uses the same 1,000 datasets in a cell; 499 row draws are nested in 1,999. FWER equals input undercoverage in this all-ties population. Intervals are individual Wilson Monte Carlo intervals.

p, n	Calibration, studentizer, B	FWER	95% MC interval
12, 252	Row, bootstrap SD, 499	0.098	[0.081, 0.118]
12, 252	Row, bootstrap SD, 1999	0.088	[0.072, 0.107]
12, 252	Row, influence SD, 1999	0.085	[0.069, 0.104]
12, 252	Gaussian, influence SD, 1999	0.097	[0.080, 0.117]
52, 252	Row, bootstrap SD, 499	0.215	[0.191, 0.242]
52, 252	Row, bootstrap SD, 1999	0.162	[0.140, 0.186]
52, 252	Row, influence SD, 1999	0.152	[0.131, 0.176]
52, 252	Gaussian, influence SD, 1999	0.188	[0.165, 0.213]
52, 1008	Row, bootstrap SD, 499	0.107	[0.089, 0.128]
52, 1008	Row, bootstrap SD, 1999	0.086	[0.070, 0.105]
52, 1008	Row, influence SD, 1999	0.067	[0.053, 0.084]
52, 1008	Gaussian, influence SD, 1999	0.076	[0.061, 0.094]

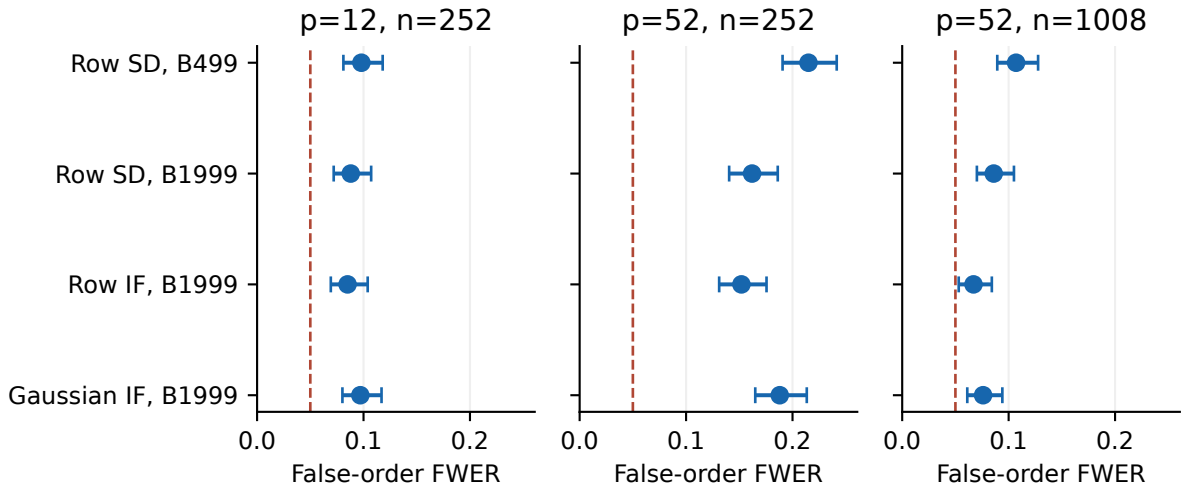


Figure 2: False-order FWER in the paired IID audit. Horizontal intervals are individual Wilson Monte Carlo intervals; the vertical line marks 0.05.

9 Cross-sectional morphometric applications

9.1 Recorded measurements and sampling scope

The primary application uses all 4,177 specimens in the UCI Abalone dataset (Nash et al., 1994). Vertices are its eight numerical measurements: length, diameter, height, whole weight, shucked weight, viscera weight, shell weight and ring count. The recorded sex/status field defines three additional reported strata (female, male and infant). Original supplied measurements are retained, including zero heights and tied ring counts; no outlier removal, imputation or residual fitting is performed. The supplied complete-case dataset has no identical full rows. One independently generated permutation fixes the observation ordering before disjoint pairing.

The sampling model treats specimens as independent observations from one common distribution within each analysed population. This is a working model for cross-sectional measurements. The repository does not provide specimen identifiers, sampling clusters or acquisition batches sufficient to establish independence from the files alone. Pooled and sex-stratified results are all reported, and heterogeneity across strata is examined explicitly. The finite-sample guarantee is conditional on this IID sampling model. The nominal error level applies within one stated method and comparison family, not jointly across all reported analyses.

As a secondary computational stress study, we apply all methods to the 16 image-derived features of UCI Dry Bean, pooled and separately for all seven varieties (Köklü and Özkan, 2020). This file contains 68 repeated full rows and several mechanically related shape measurements. We retain them and report ties and degenerate contrasts. These cases assess projection and tie handling; their sampling design is insufficiently documented for an unconditional population confidence interpretation. Morphometric correlation trees describe association among measurements, including measurement identities, rather than causal or conditional-independence networks.

9.2 Exact, merged and multiplier outputs

All eight methods and both comparison families are applied to all 12 datasets. Every projection includes all degree and ordered degree-difference LP objectives, with independent rational verification of improvements. Ordinary row-bootstrap Kendall trees, using 499 resamples and the complete tau-a statistic, supply edge frequencies and unique-hub frequencies on the same data. These frequencies remain separate from the exact confidence statements. Original datasets are redistributed under their CC BY 4.0 licences, with source citations and unchanged-file hashes.

Table 7: Primary Abalone application, $n = 4177, p = 8$, local family. P, M8, E8, E64, U-H and B-H have finite-sample guarantees under the stated IID sampling model; M-C and M-B use asymptotic multiplier calibration. All methods receive the same full degree-pair path projection.

Test	Orders	Forced	Fixed degrees	Hubs	W_d minor	W_d path	W_r
P	104	0	1	0	3.250	2.625	6.750
M8	89	0	0	0	4.250	3.250	7.000
E8	109	3	1	0	2.125	1.875	6.500
E64	111	3	2	0	1.875	1.375	5.625
U-H	42	0	0	0	6.000	5.500	7.000
B-H	45	0	0	0	6.000	5.250	7.000
M-C	152	6	6	1	0.250	0.250	3.250
M-B	125	4	3	1	0.875	0.750	4.875

For the single-matching exact reference, path projection reduces mean degree width from 3.250 to 2.625 (19.2%). Ring count has degree [1,1] and rank [3,8], excluding hub status under the stated model. Whole weight appears as the unique sample hub in all 499 ordinary Kendall bootstrap trees, but its one-matching exact rank remains [1,8]. The multiplier-contrast projection

instead gives whole-weight degree [4,4] and rank [1,1]. This nominal bootstrap declaration and the exact confidence output are reported separately. The estimated Pearson and Kendall Abalone trees share 5 of seven edges. Both weight definitions and all edge frequencies are archived. Their agreement in this dataset does not make the population targets interchangeable outside models with a common monotone correlation ordering. Pooled and sex-stratified results, every complete-family comparison, and all Dry Bean stress outputs are supplied without selecting favorable strata. The exact E64 construction gives whole-weight degree [3,5] and rank [1,3], excluding 4 of the eight vertices from hub status. Its mean degree width is 1.375, 47.6% below the single-matching exact result after the same projection. It retains a finite-sample guarantee under the same sampling model; unique-hub identification and a top-three rank restriction are distinct conclusions.

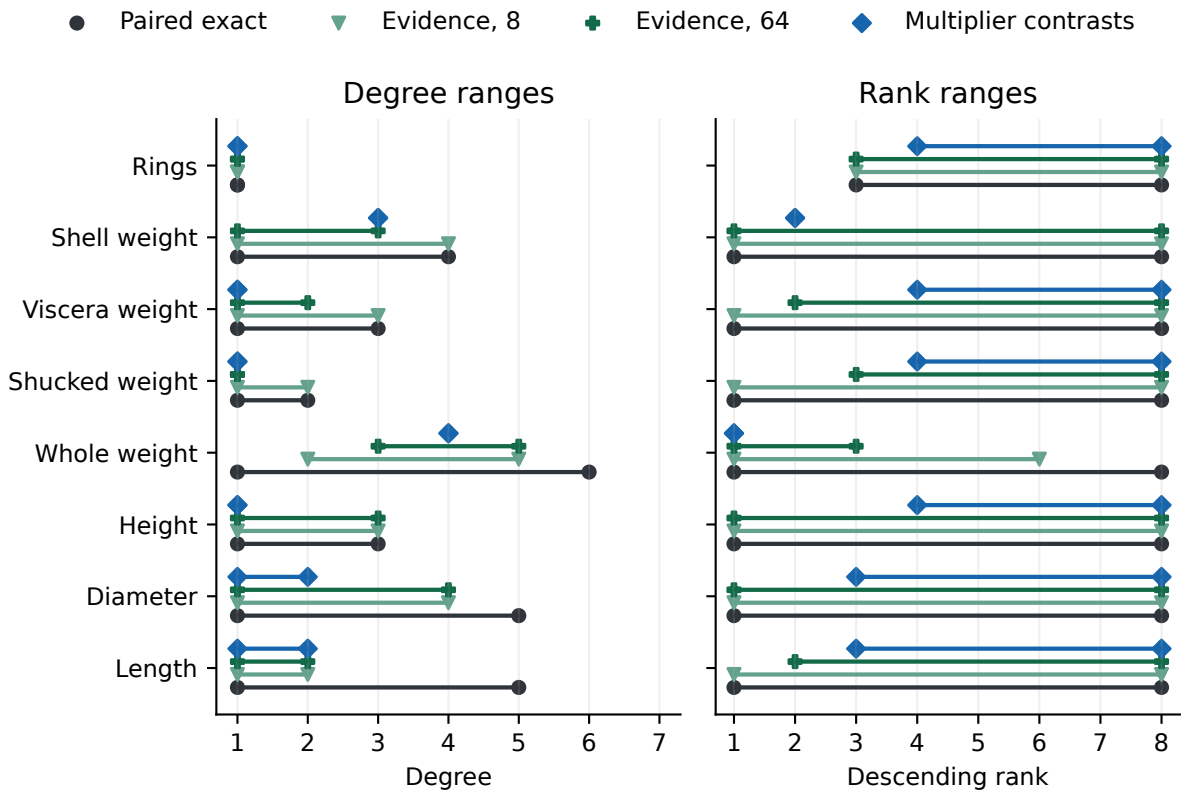


Figure 3: All eight Abalone degree and rank ranges after identical path projection. P, E8 and E64 have finite-sample guarantees under the stated IID model; M-C has asymptotic multiplier calibration. Population degree ties retain their full rank ranges.

10 Polish equity sensitivity study

Polish stock correlation trees motivate the financial setting (Gałązka, 2011). A frozen historical assembly combines publicly available Bankier price histories, WIG observations and archived GPW index portfolios. The base panel contains 231 origins from 2007–2025, each with 252 daily returns and 41–60 eligible stocks. Complete-history, positive-volume and nonzero-return checks are retained. Vendor prices are not certified dividend-inclusive total returns, extraordinary membership changes are not fully reconciled, and missing historical series prevent a complete licensed-market reconstruction.

Three representations—raw returns, cross-sectional demeaning and WIG residualization—use the same 999 circular block-bootstrap draws with block length ten, motivated by dependent

resampling (Künsch, 1989), giving 693 base analyses. A fixed annual grid adds 466 analyses with nested liquid-stock groups, 252/504/1008-session lookbacks and block-length checks. These target Pearson trees and serially dependent returns. Their interpretation is sensitivity and temporal stability, separate from the exact Kendall IID applications above. Weekly or monthly nonoverlapping returns would not by themselves establish IID sampling.

Across matched stock groups, increasing the lookback from 252 to 1008 sessions reduces mean degree width by 13.9%, 14.1% and 10.3% for groups of 10, 20 and 40 stocks. All rank ranges remain $[1, p]$. All 81 exclusions from the planned primary grid and 51 unavailable primary forward configurations per representation are retained. Full panel results, forward summaries and independent checks of 109,548 improving rational certificates appear in the supplement. This case study identifies a resolution limitation; it does not validate nominal bootstrap coverage for financial time series.

11 Computational verification and limitations

The implementation checks 300 random graphic minors against exhaustive enumeration of all labelled trees for $p = 3, 4, 5, 6$. All individual degree and pairwise degree extrema agree. The same 300 instances check exact rank projection over every feasible degree vector. In 32 additional examples with tied integer edge weights, exhaustive optimal trees satisfy every inserted path constraint and every reported LP bound. Independent rational arithmetic verifies 242 improved small-instance certificates. Thirty tied and continuous samples check the accelerated full-sample Kendall tau-a against the SciPy-based reference, and 30 samples check the optimized discordance counts against direct indicator counts. Exact-binomial superuniformity is also checked numerically over 196 combinations of sample size, null probability and significance threshold. These checks verify the implementation of the proved constructions.

For the Polish-market sensitivity study, independent Python-integer verification checks 85,129 improving rational certificates in the 693 base origin–representation analyses, plus 24,419 certificates in the 466 extended analyses. The combined GPW total is 109,548 certificates across 1,159 analyses. The base and extended scopes are reported separately in the appendix. Source-input hashes, fixed seeds, full replication records, derived order relations and the verifiers are retained. Raw third-party price series are not required to check the saved optimization certificates.

A GPT-6-based ChatGPT/Codex assistant (OpenAI) supported programming, data retrieval and extraction, exploration of statistical constructions, and LaTeX editing during this revision. Its research use included proposing the matching-evidence formulation and proof drafts. The documented verification combines direct-kernel identities, exhaustive small-tree checks, fixed-seed simulations and independent integer certificate checks, as described above.

The computational scope is explicit. The partial-order minor and path LP are outer families, and the conservation endpoints are sharp within their integer degree relaxation. A complete edge family costs $O(p^4)$ comparisons, whereas the local family costs $O(p^3)$. Fenwick sweeps compute full Kendall influences in $O(p^2 n \log n)$ time; the straightforward pairwise minor calculations visit $O(p^2 m)$ edges up to union–find factors. The flow LP has $3m$ variables plus its path constraints. Full ordered degree projection solves $2p + p(p-1)$ objectives, and anchor projection solves $2p + 2(p-1)$. Rational certificates verify numerical bounds independently of floating-point optimality checks.

12 Discussion

Direct statistical comparisons and joint tree projection address different losses of information. Averaged matching evidence retains finite-sample control under IID sampling and ties and is consistent for each fixed positive gap. Its gain over one-matching McNemar inference is

measured alongside the information cost of disjoint pairing. Full-U-statistic comparators assess how much resolution is available when all observation pairs are used. Matched projections and comparison-family ablations measure computational gains separately from testing gains.

The morphometric example makes the distinction tangible: high bootstrap hub frequency coexists with broad exact hub uncertainty, yet exact path projection can determine a leaf degree and exclude hub status. Multiplier results answer the same Kendall-tree question under asymptotic calibration. The Pearson audit compares particular implementations and finite-sample designs; its findings concern those procedures, with serial dependence requiring separate justification. The Polish panel contributes sensitivity evidence for that setting.

Useful inference need not recover the entire tree or force a unique hub. Degree restrictions, strict vertex comparisons and exclusions from a top-rank set can themselves be informative. Their credibility depends jointly on the sampling model, simultaneous input calibration and a valid projection. The framework supplies a reproducible way to inspect each of these stages.

A Complete Polish-market sensitivity summaries

Table 8: All 231 windows per representation. Widths are averages of the within-window mean degree width. Full ranks is the mean fraction of vertices with interval $[1, p]$. Hubs counts certified unique hubs over all windows. These are nominal bootstrap diagnostics.

Representation	Minor width	Path-LP width	Full ranks	Hubs
Raw	49.89	45.99	1.00	0
Demeaned	49.90	46.51	1.00	0
WIG residual	49.91	47.63	1.00	0

The refined projection reduces the raw-network mean degree width from 49.89 to 45.99, a 7.8% reduction relative to the minor. It does not resolve hub ranks: all 35,979 reported vertex–window–representation rank intervals remain $[1, p]$. No unique hub and no top-decile membership is certified. There are 85,129 saved rational certificates for improved degree or anchor-contrast endpoints. These improvements demonstrate a stricter feasible-set projection, but not an informative financial hub ranking.

The complete panel leaves financial hub ranks unresolved under the stated Pearson calibrations and outer projections. The archived 24-cell diagnostic and paired IID audit document calibration limitations; these market ranges are interpreted as sensitivity outputs.

A.1 Longer windows, nested liquid universes and forward diagnostics

The full-panel result motivates a sensitivity analysis that separates sample length from changing stock composition. Before computing the extended outcomes, we fixed the last recorded origin in every represented calendar year, giving 19 annual origins. The complete retrospective sensitivity grid and its unavailable cases are reported. At each origin, we use the longest calendar-available lookback from 504 and 1008 sessions to screen the original eligible list. Eligibility requires complete past returns, supplied prices and volumes, positive volume on at least 90% of sessions, at most 15% zero returns, and positive variance. Within this eligible set we rank stocks by median supplied closing price times volume, breaking ties by ISIN. This is an available-data liquidity proxy, not a certified corporate-action-adjusted traded-value measure.

We take nested groups of 10, 20 and 40 stocks. Their membership is kept fixed when comparing 252, 504 and 1008 sessions at the same origin. Changing group size changes the population tree target; a hub of a small liquid subnetwork need not be a hub of the full market. Demeaning is recomputed within each selected group, while factor residualization retains the same WIG series. All dated observations used for selection precede or coincide with the origin;

future observations never enter selection. Vendor histories can incorporate later revisions, so this rule does not establish a fully point-in-time data source.

Of 513 planned origin–lookback–group–representation combinations, 81 cannot be evaluated because of insufficient calendar history or too few eligible stocks. We retain the full exclusion list. The remaining 432 primary analyses use block length ten and 999 common-row resamples per origin and lookback, across all three representations. A further 34 raw-network analyses examine block lengths five and twenty at $p = 20, n = 1008$, for all 17 available origins. These are nominal, per-configuration bootstrap projections. We do not claim simultaneous coverage over the grid or valid 95% coverage for dependent returns.

Table 9: Raw-network comparison on identical stocks and matched origins. The 1008-session availability criterion leaves 17 origins for groups of 10 and 20, and ten for groups of 40. Widths and forced-edge counts are within-group origin averages. Full ranks is the fraction of vertex ranges remaining $[1, p]$.

p	Origins	n	Degree width	Forced edges	Full ranks
10	17	252	6.71	0.47	1.00
10	17	504	6.18	0.59	1.00
10	17	1008	5.78	1.00	1.00
20	17	252	15.04	0.47	1.00
20	17	504	14.14	0.65	1.00
20	17	1008	12.92	1.35	1.00
40	10	252	32.82	0.00	1.00
40	10	504	31.03	0.20	1.00
40	10	1008	29.45	0.80	1.00

The matched-origin mean degree width decreases by 13.9%, 14.1% and 10.3% when moving from 252 to 1008 observations for groups of 10, 20 and 40 respectively. More edges are forced under the nominal order region, but none of the 432 primary or 34 block-sensitivity configurations yields a non-full vertex rank interval or a certified unique hub. The negative ranking result therefore survives a fourfold lookback extension, liquidity-based dimension reduction, and the stated block checks. This describes resolution for the stated procedures and targets.

Table 10: Block-length sensitivity for the same 17 raw-network origins, $p = 20, n = 1008$. Resampling seeds differ between block lengths.

Block length	Origins	Degree width	Forced edges	Full ranks
5	17	12.94	1.24	1.00
10	17	12.92	1.35	1.00
20	17	12.93	1.29	1.00

For a forward diagnostic, we hold the training stock group fixed and estimate a new tree using the next 252 sessions. The diagnostic is available only when every selected stock and the factor have complete future returns. We neither impute unavailable histories nor drop a future-missing stock from the selected group. There are 93 complete primary configurations per representation, out of 144; the remaining 51 have incomplete future histories or insufficient follow-up. Overlapping lookbacks and nested groups mean these are not 93 independent samples. Forward summaries are conditional on future-history availability and may be affected by attrition; no reweighting corrects that selection. We compute the fraction of transitive training order statements whose direction agrees with the future correlation estimates, and whether the training anchor remains tied for largest future degree. These are descriptive temporal-stability measures, not population coverage tests or portfolio-performance results.

Estimated edge-order directions persist much more frequently than the identity of the

Table 11: Forward diagnostics over the next 252 sessions. Order agreement averages the within-configuration fractions, giving each configuration equal weight. Anchor still top includes ties and does not establish a unique hub.

Representation	Configurations	Order agreement	Anchor still top
Raw	93	95.7%	35/93
Demeaned	93	95.2%	19/93
WIG residual	93	93.9%	19/93

estimated highest-degree vertex. This comparison involves different functionals: preservation of many inequalities is not equivalent to preservation of a hub, and a high fraction of correct directions need not mean that every direction is preserved jointly. Because the population can change over time, the forward sample is not a known-truth coverage benchmark. The calibration failures in Section 7 remain unresolved.

As an illustration, the earliest complete raw-network configuration with anchor bootstrap frequency at least 0.9 and loss of top position in the future sample is the 2008 annual origin, $p = 10, n = 504$. The anchor is PKOBP; it is tied for highest degree in 91.2% of bootstrap trees, while its nominal degree range is [1, 9] and rank range is [1, 10]. Its future estimated rank range is [3, 10], despite 100.0% directional agreement for the certified training orders. The illustration is selected retrospectively by the stated rule; all configurations are reported, and it is not additional confirmatory evidence.

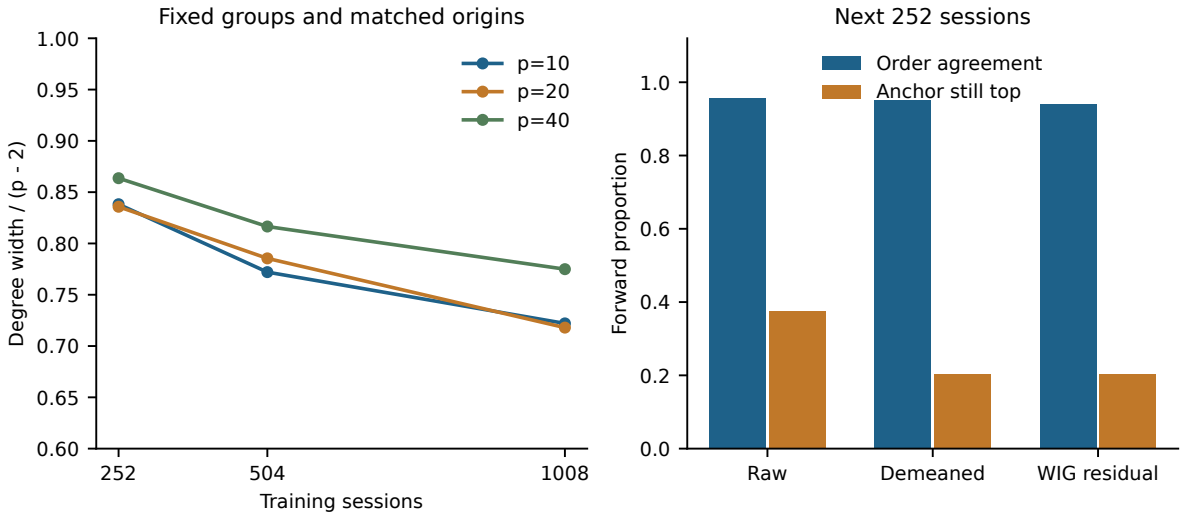


Figure 4: Extended empirical diagnostics. Left: mean raw-network degree width divided by $p-2$, on the matched origins and fixed stock groups in Table 9. Right: forward edge-order agreement and retention of the training anchor among the highest-degree vertices, from Table 11. Both panels are descriptive; neither supplies a validated confidence level for GPW returns.

Independent verification reconstructs the selection rule for all 50 origin–group universes, checks five source-input hashes and all 9,710 additional vertex records, recomputes the 303 available forward analyses including block sensitivities, and validates all 24,419 improving rational LP certificates. Every check passes. Longer financial windows increase sample size while raising the possibility of regime changes, so their improved degree resolution must be interpreted alongside that stationarity limitation.

Funding

No dedicated funding was received for this study.

Competing interests

The authors declare no relevant competing interests.

Data availability

Derived results, fixed simulation designs, data provenance and verification code accompany the article. The empirical analysis uses the frozen historical data assembly described above. Original UCI measurement files and their CC BY 4.0 attribution are included. Raw third-party stock-price histories are not redistributed; source records and hashes document their availability and limitations.

Code availability

Reproducibility code and immutable identifiers are supplied separately to the editors.

Declaration of generative AI and AI-assisted technologies in manuscript preparation

The authors used ChatGPT/Codex (OpenAI) to assist programming, data handling, exploration of methodological formulations and manuscript editing. The authors reviewed and validated the outputs and take responsibility for the article. Research-related assistance and verification procedures are described in the computational methods.

References

Xiaohui Chen. Gaussian and bootstrap approximations for high-dimensional U-statistics and their applications. *The Annals of Statistics*, 46(2):642–678, 2018. doi: 10.1214/17-AOS1563.

Xiaohui Chen and Kengo Kato. Randomized incomplete U-statistics in high dimensions. *The Annals of Statistics*, 47(6):3127–3156, 2019. doi: 10.1214/18-AOS1773.

Victor Chernozhukov, Denis Chetverikov, and Kengo Kato. Gaussian approximations and multiplier bootstrap for maxima of sums of high-dimensional random vectors. *The Annals of Statistics*, 41(6):2786–2819, 2013. doi: 10.1214/13-AOS1161.

Jérôme Fortin, Adam Kasperski, and Paweł Zieliński. Some methods for evaluating the optimality of elements in matroids with ill-known weights. *Fuzzy Sets and Systems*, 160:1341–1354, 2009. doi: 10.1016/j.fss.2008.11.013.

Marek Gałązka. Characteristics of the Polish Stock Market correlations. *International Review of Financial Analysis*, 20(1):1–5, 2011. doi: 10.1016/j.irfa.2010.11.002.

Peter Hall and Hugh Miller. Using the bootstrap to quantify the authority of an empirical ranking. *The Annals of Statistics*, 37(6B):3929–3959, 2009. doi: 10.1214/09-AOS699.

Wassily Hoeffding. Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association*, 58(301):13–30, 1963. doi: 10.1080/01621459.1963.10500830.

Sture Holm. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70, 1979. URL https://www.jstor.org/stable/4615733.

Adam Kasperski and Paweł Zieliński. On combinatorial optimization problems on matroids with uncertain weights. *European Journal of Operational Research*, 177(2):851–864, 2007. doi: 10.1016/j.ejor.2005.12.033.

Murat Köklü and Ilker Ali Özkan. Dry bean, 2020. URL https://archive.ics.uci.edu/dataset/602/dry+bean+dataset. Dataset, UCI Machine Learning Repository.

Hans R. Künsch. The jackknife and the bootstrap for general stationary observations. *The Annals of Statistics*, 17(3):1217–1241, 1989. doi: 10.1214/aos/1176347265.

Quinn McNemar. Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika*, 12:153–157, 1947. doi: 10.1007/BF02295996.

Magne Mogstad, Joseph P. Romano, Azeem M. Shaikh, and Daniel Wilhelm. Inference for ranks with applications to mobility across neighbourhoods and academic achievement across countries. *The Review of Economic Studies*, 91(1):476–518, 2024. doi: 10.1093/restud/rdad006.

Federico Musciotto, Luca Marotta, Salvatore Miccichè, and Rosario N. Mantegna. Bootstrap validation of links of a minimum spanning tree. *Physica A: Statistical Mechanics and its Applications*, 512:1032–1043, 2018. doi: 10.1016/j.physa.2018.08.020.

Warwick Nash, Tracy Sellers, Simon Talbot, Andrew Cawthorn, and Wes Ford. Abalone, 1994. URL https://archive.ics.uci.edu/dataset/1/abalone. Dataset, UCI Machine Learning Repository.

Matey Neykov, Junwei Lu, and Han Liu. Combinatorial inference for graphical models. *The Annals of Statistics*, 47(2):795–827, 2019. doi: 10.1214/17-AOS1650.

Seth Pettie. Sensitivity analysis of minimum spanning trees in sub-inverse-ackermann time, 2014. URL https://arxiv.org/abs/1407.1910. arXiv:1407.1910.

Michele Tumminello, Claudia Coronnello, Fabrizio Lillo, Salvatore Miccichè, and Rosario N. Mantegna. Spanning trees and bootstrap reliability estimation in correlation-based networks. *International Journal of Bifurcation and Chaos*, 17(7):2319–2329, 2007. doi: 10.1142/S0218127407018415.

Vladimir Vovk and Ruodu Wang. Combining p-values via averaging. *Biometrika*, 107(4):791–808, 2020. doi: 10.1093/biomet/asaa027.

Vladimir Vovk and Ruodu Wang. E-values: calibration, combination and applications. *The Annals of Statistics*, 49(3):1736–1754, 2021. doi: 10.1214/20-AOS2020.

Hande Yaman, Oya Ekin Karaşan, and Mustafa Ç. Pınar. The robust spanning tree problem with interval data. *Operations Research Letters*, 29(1):31–40, 2001. doi: 10.1016/S0167-6377(01)00078-5.

Marco Zaffalon and Marcus Hutter. Robust inference of trees. *Annals of Mathematics and Artificial Intelligence*, 45:215–239, 2005. doi: 10.1007/s10472-005-9007-9.